

Microprocessors and Microcontrollers

Assignment 7- Week 7

TYPE OF QUESTION: MCQ

Number of questions: 14

Total mark: 14 X 1 = 14 M

QUESTION 1:

Correct Answer: a

Detailed Solution:

The **8051 microcontroller** has **four register banks (Bank 0 to Bank 3)** for registers **R0 to R7**. When the **8051 is powered up**, it defaults to **Register Bank 0** because the **RS0 and RS1 bits** in the **PSW (Program Status Word) register** are both cleared (**RS1 = 0, RS0 = 0**).

To switch to other register banks, the **RS0 and RS1 bits** in the **PSW register** must be modified.

Register Bank	RS1 (PSW.4)	RS0 (PSW.3)	Address Range
Bank 0 (Default)	0	0	00H - 07H
Bank 1	0	1	08H - 0FH
Bank 2	1	0	10H - 17H
Bank 3	1	1	18H - 1FH

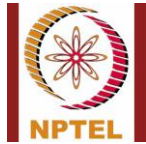
Since **Bank 0 is the default bank after a reset**, option (a) **Bank 0** is the correct answer.

QUESTION 2:

Correct Answer: b

Detailed Solution:

The **8051 microcontroller** has **four register banks (Bank 0 to Bank 3)** for registers **R0 to R7**. The selection of a register bank is controlled by **RS1 (PSW.4)** and **RS0 (PSW.3)** bits in the **Program Status Word (PSW) register** as follows:



Register Bank RS1 (PSW.4) RS0 (PSW.3) Address Range

Bank 0 (Default)	0	0	00H - 07H
Bank 1	0	1	08H - 0FH
Bank 2	1	0	10H - 17H
Bank 3	1	1	18H - 1FH

Step-by-step analysis of the instruction:

- The instruction **SETB PSW.3** sets **PSW.3 (RS0) = 1**, while **PSW.4 (RS1) remains 0** (since it was not modified).
- This results in **RS1 = 0** and **RS0 = 1**, which corresponds to **Register Bank 1**.

QUESTION 3:

Correct Answer: (d)

Detailed Solution:

The 8051 microcontroller has an 8-bit Stack Pointer (SP), which determines the location where data is stored when using the stack. The following statements clarify each option:

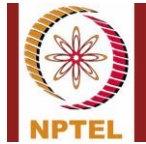
Option A: TRUE

The stack pointer in the 8051 is only eight bits wide.

- The SP register is 8 bits wide, meaning it can address memory locations from 00H to FFH.

Option B: TRUE

When the 8051 is powered up, the SP register contains the value of 07H.



- Upon reset, the SP is initialized to 07H, meaning the stack starts from 08H when data is pushed.

Option C: TRUE

The storing of a CPU register in the stack is called PUSH.

- The PUSH instruction saves the contents of a register or memory location onto the stack.

Option D: FALSE (Correct Answer)

As we push data onto the stack, the stack pointer is decremented by 1.

- This statement is incorrect because in the 8051, the stack pointer is incremented (not decremented) before storing data when using the PUSH instruction.

Correct Behavior of Stack Pointer (SP) in 8051:

- PUSH instruction → SP is incremented first, then data is stored.
- POP instruction → Data is retrieved first, then SP is decremented.

QUESTION 4:

Correct Answer: d

Detailed Solution:

Step 1: Initializing the Stack Pointer

MOV SP, #6FH

- This sets SP = 6FH initially.

Step 2: Register Assignments (No effect on Stack)

MOV R1, #10H ; Loads 10H into R1

MOV R6, #0FH ; Loads 0FH into R6



MOV R3, #03H ; Loads 03H into R3
MOV R2, #FFH ; Loads FFH into R2

- These instructions do not affect the stack.

Step 3: PUSH Instructions

First PUSH: PUSH 3

- R3 = 03H, so we push 03H onto the stack.
- SP increments to 70H, and (70H) = 03H.

Second PUSH: PUSH 2

- R2 = FFH, so we push FFH onto the stack.
- SP increments to 71H, and (71H) = FFH.

Third PUSH: PUSH 1

- R1 = 10H, so we push 10H onto the stack.
- SP increments to 72H, and (72H) = 10H.

Fourth PUSH: PUSH 6

- R6 = 0FH, so we push 0FH onto the stack.
- SP increments to 73H, and (73H) = 0FH

Final Stack State

Stack Pointer (SP) = 73H

Top of the stack = 0FH (stored at address 73H)



QUESTION 5:

How many times is the instruction **CPL A** executed in the following **8051** assembly program?

MOV A, #55H

MOV R2, #50H

START: MOV R5, #0CH

LOOP: CPL A

DJNZ R5, LOOP

DJNZ R2, START

- a) **600 times**
- b) **800 times**
- c) **900 times**
- d) **960 times**

Correct Answer: d

Detailed Solution:

Step 1: Initialization

MOV A, #55H ; Load A with 55H (irrelevant for count)

MOV R2, #50H ; Load R2 with 50H (R2 = 80 in decimal)

- R2 = 80 (Outer loop counter)

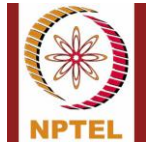
Step 2: Outer Loop (**START**)

START: MOV R5, #0CH ; Load R5 with 0CH (R5 = 12 in decimal)

Step 3: Inner Loop (**LOOP**)

LOOP: CPL A ; Complements A

DJNZ R5, LOOP ; Decrement R5 and loop back if not zero



- CPL A executes 12 times per outer loop iteration.

Step 4: Outer Loop Iteration

DJNZ R2, START ; Decrement R2 and loop back if not zero

The outer loop runs 80 times (R2 starts at 80 and decrements to 0).

Each time, CPL A executes 12 times.

Step 5: Total Executions of **CPL A**

$80 \times 12 = 960$

QUESTION 6:

What is the content of Register A after executing following instructions?

```
MOV A,#0
MOV R2,#10
AGAIN: ADD A,#03
       DJNZ R2,AGAIN
       MOV R5,A
```

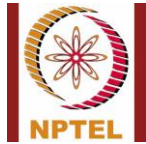
- a) A = 10
- b) A = 13
- c) A = 30
- d) A = 0

Correct Answer: (c)

Detailed Solution:

1. Initial Values:
 - A = 0
 - R2 = 10
2. Loop Execution (AGAIN label)
 - Instruction ADD A, #03 runs 10 times (since R2 starts at 10 and decrements to 0).
 - Each iteration adds 3 to A.

$$A = 0 + (10 \times 3) = 30$$



QUESTION 7:

In the **8051 microcontroller**, what are the **rollover values** for the timers in **Mode 2** and **Mode 3**?

- a) **00FFH, FFFFH**
- b) **1FFFH, 0FFFH**
- c) **00FFH, FFFFH**
- d) **0FFH, FFFFH**

Correct Answer: (d)

Detailed Solution:

Understanding Timer Modes in 8051:

1. Mode 2 (8-bit Auto-Reload Timer)
 - This mode uses an 8-bit counter, meaning it counts from 00H to FFH (255 in decimal) before rolling over.
 - Rollover value = 0FFH
2. Mode 3 (Split Timer Mode - Two 8-bit Timers in Timer 0)
 - If Timer 0 is used in Mode 3, it splits into two 8-bit timers that count independently.
 - Timer 1 still functions as a full 16-bit timer, meaning it rolls over at FFFFH.
 - Rollover value = FFFFH

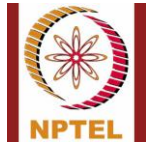
QUESTION 8:

Correct Answer: (d)

Detailed Solution:

The 8051 microcontroller generates multiple control signals to access external memory and devices:

- ALE (Address Latch Enable) – Used to latch the lower byte of the address during external memory access.
- PSEN (Program Store Enable) – Used to enable external program memory.



- RD & WR (Read and Write signals) – Used for reading from and writing to external data memory.

QUESTION 9:

Correct Answer: a

Detailed Solution:

QUESTION 10:

Correct Answer: (c)

Detailed Solution:

QUESTION 11:

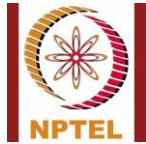
Which instruction can be used to select Register Bank 3 in the 8051 microcontroller?

- a) SETB PSW.3
- b) SETB PSW.4
- c) SETB PSW.3 and SETB PSW.4
- d) CLR PSW.3 and CLR PSW.4

Correct Answer: (c)

Detailed Solution:

The register bank selection in 8051 is controlled by RS1 (PSW.4) and RS0 (PSW.3).
To select Bank 3, both RS1 and RS0 must be set to 1.
SETB PSW.3 sets RS0 = 1.
SETB PSW.4 sets RS1 = 1.
Since RS1 = 1 and RS0 = 1, Register Bank 3 is selected.



QUESTION 12:

Which control signal is used to latch the lower byte of an address while accessing external memory in the 8051?

- a) PSEN
- b) RD
- c) ALE
- d) WR

Correct Answer: (c)

Detailed Solution:

The **Address Latch Enable (ALE)** signal is used to latch the lower 8 bits of the address during external memory access.

The 8051 multiplexes the lower byte of the address with data, so ALE helps separate them.

QUESTION 13:

What is the primary advantage of RISC architecture?

- a) More instructions per cycle
- b) Simpler instruction set, faster execution
- c) Complex addressing modes
- d) More power consumption

Correct Answer: (b)

Detailed Solution:



QUESTION 14:

If the Stack Pointer (SP) is initialized to 50H, what will be its value after executing the following instructions?

PUSH A
PUSH B
PUSH R2
PUSH R7

- a) 54H
- b) 56H
- c) 52H
- d) 58H

Correct Answer: (a)

Detailed Solution:

Initially, SP = 50H.

Each PUSH instruction increments SP by 1 before storing the data.

First PUSH → SP = 51H

Second PUSH → SP = 52H

Third PUSH → SP = 53H

Fourth PUSH → SP = 54H

*****END*****